Inorganic CUME

April-19-2013

The newly established "Energy Storage and Battery Hub" led by Argonne National Lab under a $200 million grant, the highest ever from the US-DOE, implies that "energy storage and battery" will be a hot topic in research, economy, and job market for decades to come. The seminar given by Dr. Cao on March-29-2013 reemphasized the importance of materials chemistry in the energy science and economy.

(1). Pick a rechargeable battery of your choice, and briefly discuss the inorganic reactions on the anode and cathode during the battery's charging/discharging processes. (**25'**)

(2). Manganese oxides are commonly used as battery materials. Briefly define the EMF. Indicate the sign (positive or negative) of each EMF in the following diagram and briefly explain who so, based on the fact that the oxidation states (IV) and (II) being more stable than the rest. (**25'**)

$$Mn^{4+} \xrightarrow{\text{EMF:}} Mn^{3+} \xrightarrow{\text{EMF:}} Mn^{2+} \xrightarrow{\text{EMF:}} Mn^{0}$$

(3). A low-cost rechargeable battery in most cars uses too much of an element that is heavy and toxic. Propose a new electrode material for largely replacing the toxic element in this type of battery, and briefly discuss your inorganic chemistry for supporting your proposal. (**25'**)

(4). Propose a novel inorganic reaction (or a material) for safely and economically storing energy other than the electrical energy, and briefly discuss the working principle of the energy-storage process. (**25'**)